

Can It Ford? Query-Conditioned, Physically Viable World Models for Autonomous-Vehicle Flood Traversability from Gaussian Splatting

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Abstract—Flooded roads are a leading cause of flood fatalities, killing drivers who cannot judge how deep the water is, how fast it moves, or what the road beneath it is doing. Automated vehicles inherit the same gap: a world model predicting only appearance can render an outcome that looks plausible but is wrong. Physically viable world models preserve that latent physics, but leave open how much fidelity a query requires. This project asks a narrow question, can an autonomous vehicle ford a specific flooded road, and identifies the simplest abstraction sufficient to answer it, not the most detailed simulation available. We compare three fidelity levels on a depth-velocity grid: a static depth cutoff, the Australian Rainfall and Runoff (AR&R) criterion, and a coupled rigid-body Material Point Method simulation of a watertight Toyota Yaris hull. A real scene is reconstructed as an unscaled splat, but with no PhysGaussian bridge built, every flood domain is parametric. Across 70 conditions the depth cutoff admits 7 crossings and the joint criterion 14; tightening the rule and the class together reclassifies 23 cells to NO-FORD with none in the opposite direction, and vehicle class alone decides the verdict in 12. In a 9-condition pilot on synthetic geometry, run with SPH rather than the coupled configuration, the two agree at only 5, and all four disagreements run one way: the criterion permits where the simulation refuses. The minimal sufficient abstraction is therefore query-dependent, and this work delivers a reconstruct-to-decide pipeline instantiating query-conditioned physically viable world models.

Index Terms—physically viable world models, Gaussian splatting, Material Point Method, flood traversability, autonomous vehicles, abstraction selection

I. INTRODUCTION

Flooded roads kill more people in the United States each year than any other flood-related hazard, and most of those deaths involve a vehicle driven into water whose depth, flow velocity, or submerged road condition could not be judged from the surface [1]. This is not only a human-driver problem. An autonomous vehicle facing the same scene has the same information gap, and a world model that predicts only visual

appearance can render a plausible-looking outcome that is physically wrong: it may show a vehicle continuing forward with no indication that the same maneuver would cause loss of traction or flotation.

Physically viable world models (PVWM) address this by preserving the latent physics that governs an action’s outcome rather than only its appearance [2]. The PVWM framework identifies automatic abstraction selection, choosing how much physical fidelity a given query actually requires, as a central open problem, and leaves the orchestrator that would make that choice unimplemented [2].

This project instantiates that orchestrator for one deliberately narrow query: *can this specific autonomous vehicle ford this specific flooded road right now?* Rather than building the highest-fidelity simulation possible, we identify the simplest physical abstraction sufficient to answer that query correctly, and characterize where a cheaper abstraction silently gives the wrong answer. Figure 1 summarizes the full reconstruct-to-decide pipeline. The Gaussian splatting front end has reconstructed and trained a splat of one real scene, but that splat is not yet metrically scaled or coupled to any result reported here, so the flood domain in every result below is specified parametrically in depth and velocity rather than recovered from reconstruction.

II. PRIOR WORK

A. Flood-Vehicle Stability Criteria

Two families of empirical criteria describe when floodwater destabilizes a vehicle. The Australian Rainfall and Runoff (AR&R) Project 10 guidelines define a depth-velocity hazard scalar, $H = D \times V$ (m^2/s), with published class-specific thresholds [3]. Those thresholds are labelled “draft interim” by the report itself, and the table they come from is titled for *stationary* vehicles, so throughout this paper L1 is treated as a documented decision rule to be tested rather than as a validated safety boundary. That stationary scope is why the comparison in this paper is well posed rather than mismatched:

This work was supported by the National Science Foundation under NSF REU Site: Cyberinfrastructure Research for Societal Advancement (SCIFE-AI), Award #2447887.

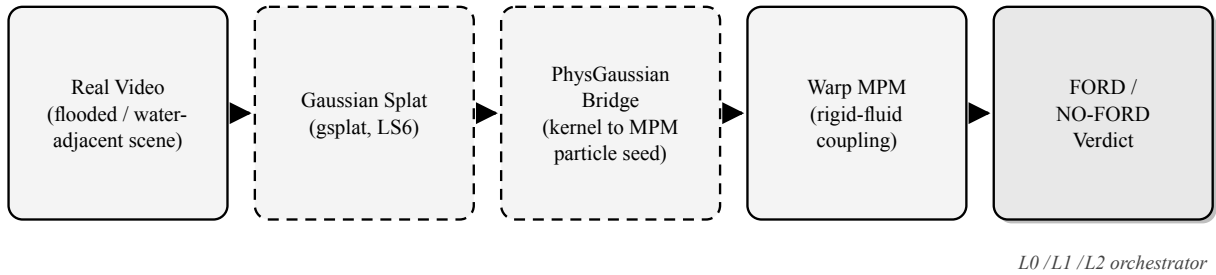


Fig. 1. The reconstruct-to-decide pipeline: a real video is reconstructed as a Gaussian splat, bridged to Material Point Method particles following PhysGaussian, simulated with rigid-fluid coupling, and resolved to a FORD/NO-FORD verdict by the L0/L1/L2 abstraction orchestrator. Dashed stages are not on the path used for any result reported here: a Gaussian splat of one real scene has been trained but is not metrically scaled, and the PhysGaussian kernel-to-particle bridge is designed and not yet built. The MPM stage names the solver that produced the coupled runs reported in Section IV-C; Genesis is the engine the full pipeline is designed around.

the coupled runs hold the vehicle as a free rigid body in a tank and drive the water past it, so both sides of the comparison describe a stationary vehicle subjected to flow. It is the word *ford* that carries the mismatch, since a real crossing is a moving vehicle, and no result here should be read as a statement about a vehicle under power. Smith, Modra, and Felder measured full-scale flotation and sliding stability curves by testing real vehicles rather than scale models [4]. Their small passenger car was a Toyota Yaris sedan, tested at full scale alongside a Nissan Patrol 4WD and a Ford Festiva, and it is the same vehicle model this paper simulates at L2 [4]. Both are single-number criteria: neither represents the direction of applied force, and neither was derived to predict the persistent lateral drift this project observes at some phase-space points (Section IV). Numerical models of vehicles in water also exist. Examples include multibody dynamics coupled to SPH for vehicle fording [5], automotive wading simulation [6], simulations of vehicle-water interaction in shallow water validated against experiments [7], SPH combined with laboratory measurements, which confirmed the AR&R criteria for stationary vehicles [8], 3D CFD of passenger vehicles exposed to water flows [9], numerical simulation of passenger vehicles moving through floodwater [10], and full-process simulation of floating vehicles in flash floods [11]. Unlike most of these, the coupled model used here has not been compared against experiments, so it serves only as the top rung of an abstraction ladder, not as an extension of that line of work in fidelity.

B. Physics-Grounded Scene Reconstruction

3D Gaussian Splatting (3DGS) reconstructs a photorealistic, explicit scene representation from ordinary multi-view video [12]. PhysGaussian extends this representation with physical simulation by treating each Gaussian kernel’s covariance as a seed for a continuum Material Point Method (MPM) particle volume, allowing a trained splat to drive a physics simulation directly rather than only a renderer [13]. The pipeline

is designed around Genesis, an open-source, solver-agnostic physics engine [14]. We are not aware of a publicly available, drop-in pipeline carrying flood video through to a Genesis MPM simulation: PhysGaussian’s own reference implementation targets a Warp MPM solver rather than Genesis, so that bridge has to be built rather than adopted, and it is not built yet.

C. Why a Single Scalar Threshold Is Insufficient

Both L0 and L1 are deterministic functions of published thresholds, not simulation output: L0 applies a single fixed depth cutoff [1], and L1 applies the AR&R two-part depth-and-hazard criterion per vehicle class [3]. We evaluate both across the same depth-velocity grid. Fig. 2 shows that L0 alone is markedly over-conservative, and that AR&R’s own published per-class caps (Fig. 3) already diverge well before any coupled simulation is involved. Neither figure required simulated data; both are direct evaluations of cited formulas, and both motivate why a coupled physical model is worth the added cost.

D. Forward and Inverse Property Estimation

Hsiao and Kumar demonstrate the inverse direction of this same reconstruction-to-physics pipeline: given only visual observation, a NeRF-based reconstruction paired with Bayesian optimization recovers a granular material’s friction angle to within roughly two degrees of ground truth [16]. This project addresses the complementary forward direction: given a known scene and known flood hydraulic conditions, what is the traversability verdict under full fluid-rigid-body dynamics? Together, the two directions describe a closed perception-to-action loop, one of the auditability properties PVWM requires [2].

III. APPROACH

A. Pipeline

The reconstruct-to-decide pipeline (Fig. 1) has four stages: (1) a real flooded or water-adjacent road scene is captured

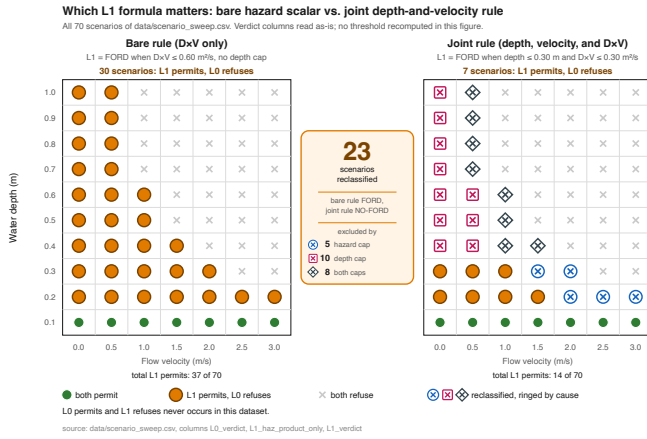


Fig. 2. L0 (fixed depth threshold) versus L1 (AR&R depth-velocity criterion) across all 70 (depth, velocity) scenarios of `data/scenario_sweep.csv`. Both panels are direct evaluations of published formulas [1], [3], not simulation output, and both read their verdicts from stored columns rather than recomputing a threshold. Left: the *bare* rule (`L1_haz_product_only`), which tests only $D \times V \leq 0.60 \text{ m}^2/\text{s}$ and permits crossing in 37 of 70 scenarios, 30 of them where L0 alone would refuse. Right: the *joint* rule (`L1_verdict`), which requires the class depth cap, the 3.0 m/s velocity cap, and the $D \times V$ cap to all hold together, permitting 14 of 70, only 7 of them beyond L0. The joint rule is canonical for this paper going forward; the bare rule is shown because it is the form in which the criterion is most often applied. The two panels differ in both rule structure and vehicle class: the left applies AR&R’s Large 4WD hazard cap alone ($D \times V \leq 0.60 \text{ m}^2/\text{s}$, no depth restriction), the right the Small Car joint rule ($D \times V \leq 0.30 \text{ m}^2/\text{s}$ and depth $\leq 0.30 \text{ m}$); `L1_verdict` is identical to `L1_verdict_small_passenger` on all 70 rows. Of the 23 scenarios that reclassify from FORD to NO-FORD between the panels, 5 are excluded by the tighter hazard cap alone, 10 by the depth cap alone, and 8 by both, and none reclassify in the opposite direction. The depth cap is not redundant with the hazard cap: 7 of the 23 are standing water at zero velocity and depths of 0.40 to 1.00 m, which a product-only threshold permits at any depth because $D \times V = 0$.

on video and reconstructed in 3D via Gaussian splatting on an NVIDIA A100 GPU (TACC Lonestar6); (2) following the PhysGaussian extraction pattern, trained Gaussian kernels are opacity-filtered and converted into an initial MPM particle volume; (3) the particle volume, together with a rigid-body vehicle proxy, is simulated with weakly-compressible MPM water and rigid-MPM coupling, in the Genesis physics engine as designed and, for the coupled runs reported here, in a Warp-based MPM solver; (4) the simulation’s displacement output is passed through a query-conditioned abstraction-selection orchestrator, `orchestrate_ford_query()`, which returns a FORD/NO-FORD verdict at the cheapest level of fidelity sufficient to answer correctly.

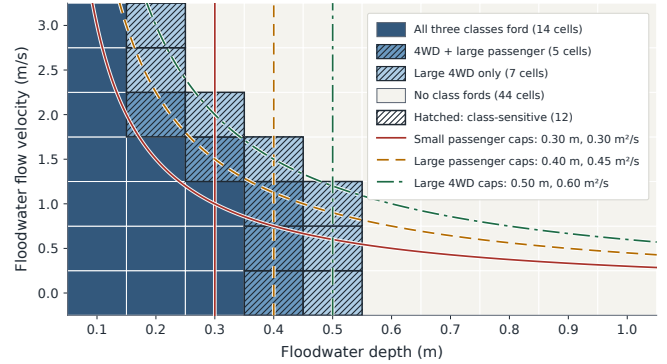
B. Three-Level Abstraction Ladder

We compare three levels of physical fidelity on the same phase-space grid of (depth, velocity) conditions, summarized in Table I.

This instantiates the orchestrator PVWM leaves unimplemented: the empirical L1/L2 divergence zone characterized in Section IV defines the escalation boundary at which L1 alone is not sufficient and the query must be answered at L2.

Vehicle class decides the verdict in 12 of 70 scenarios

L1 AR&R stationary-vehicle stability screening over a 10×7 depth and velocity grid



Caps: Shand, Cox, Blacka and Smith 2011, AR&R Project 10 Stage 2, P10/S2/020, Table 3, page 14. Draft interim values, not a safety standard. Data: `data/scenario_sweep.csv`, md5 890884346a52, 4435 bytes, 2026-07-29 18:25:47 CDT.

Fig. 3. The L1 criterion evaluated per AR&R’s own three published vehicle classes [3]: small passenger, large passenger, and large 4WD. These do not correspond to the sedan/SUV/pickup grouping used for the diagnostic vehicle asset in Table II. That grouping is ours: it is assembled from individual measured vehicle inertial parameters [15], not from any published three-class scheme. Where this paper reports results tied to AR&R’s own classes, it uses the watertight-hull mass-override method described in Section III-C, not the silhouette splat.

A first-principles force balance motivates why L1’s scalar hazard measure is expected to be insufficient. Fig. 4 plots lateral hydrodynamic drag force against maximum available static friction at several water depths (left), and the resulting critical velocity above which drag is predicted to exceed friction, as a function of depth (right). It is built from the standard quadratic drag relation, $F_D = \frac{1}{2} \rho C_d A V^2$, balanced against a Coulomb friction limit μN in which the normal force N is reduced by buoyancy. Both coefficients are pinned to measured primary sources. The drag coefficient $C_d = 1.38$ comes from Smith, Modra, and Felder’s flume testing of a 1:18 scale Toyota Yaris [4]. Azhar et al., reading that same testing, report the coefficient ranging between 1.0 and 1.8 across subcritical and supercritical flow [8]; we cite the range to them because that is where we read it. The friction coefficient $\mu = 0.30$ is the value AR&R itself adopted when deriving the L1 thresholds this figure interrogates, which Shand et al. call conservative [3]. That same 0.30 is independently the sand and gravel worst case measured by Smith, Modra and Felder, whose wet and dry concrete measurements both give 0.78 [4]; the right panel shades the 0.30 to 0.78 band. Both endpoints of that band are Smith’s measurements. AR&R adopts 0.30 as an assumption and reports no 0.78. That choice matters: the figure and the L1 criterion now rest on the same friction assumption, so divergence between them cannot be attributed to that gap. What remains unpinned is the displacement geometry, which the caption states explicitly.

C. Vehicle and Scene Representation

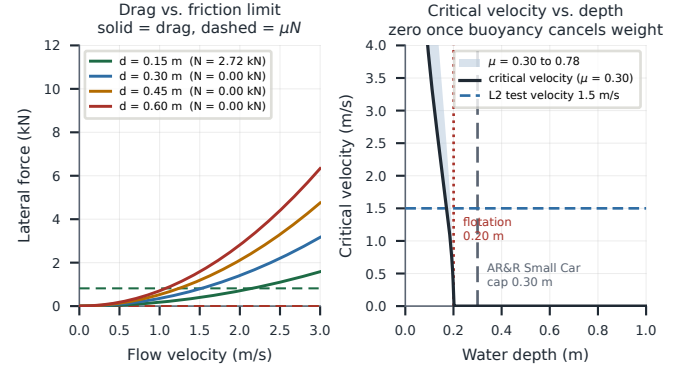
The vehicle asset used to find and root-cause this project’s grid-resolution solidification defect is a rigid body solidified from a surface Gaussian splat (`truck_trimmed.ply`,

TABLE I
ABSTRACTION LADDER DEFINITIONS

Level	Basis	Definition
L0	Depth	Fixed depth cutoff at 0.15 m, independent of velocity or vehicle class. This is the National Weather Service’s 6-inch figure, the depth at which fast-moving water can knock over an adult; the same source’s vehicle figure is twice that, 12 inches (0.30 m), at which rushing water carries away most cars [1]. We adopt the pedestrian depth deliberately, as the conservative bound of the two, and note that full-scale testing does not give one number here. A small car can lose stability in roughly 0.15 m of fast-moving water, while the still-water depth at which it floats is closer to 0.30 m [4]. A single L0 cutoff is therefore standing in for two different situations, and which one it matches depends on how fast the water is moving. Across the 70-scenario grid this admits 7 scenarios; the 0.30 m cutoff would admit 21.
L1	$D \times V$	AR&R’s joint criterion. A crossing is permitted only when all three limits hold at once: depth at or under the class cap, velocity at or under 3.0 m/s, and $D \times V$ at or under the class product cap [3]. For small passenger those are 0.30 m, 3.0 m/s, and $0.30 \text{ m}^2/\text{s}$. The velocity limit never binds on our grid, which stops at 3.0 m/s. V is flow velocity, not vehicle speed. Mass and force direction are not represented.
L2	MPM	Coupled rigid-body/MPM simulation; NO-FORD when lateral drift exceeds 0.05 m. That value is an internal numerical detector for onset of motion, set conservatively above the solver’s own displacement noise floor, and is not itself an empirical stability criterion. The sliding physics it stands in for, drag overcoming tyre-road friction on a partially submerged vehicle, is established independently [17], [18].

191,107 points, no interior), used because no watertight simulation-ready mesh existed in the pipeline at the time. Three vehicle classes (sedan, SUV, pickup) are each a single rescaled base silhouette stretched to bounding-box dimensions taken from individual measured vehicles in the NHTSA/SAE inertial-parameter database [15]. That source is a record of individually measured vehicles through November 1998; it is not a matched sedan/SUV/pickup set, and it contains no Yaris. The grouping into three classes is therefore ours, built by selecting individual entries, and carries no authority from the source beyond the per-vehicle measurements themselves. This is stated explicitly as a limitation, not glossed over: all three classes share one silhouette, so cross-class comparisons involving this asset reflect scaled mass and footprint, not distinct vehicle geometry.

Solidifying a surface point cloud into a filled rigid body is grid-resolution-dependent in a way that is easy to miss. At MPM grid resolution $n_{\text{grid}}=64$, the column-fill solidification step correctly produces a solid body and the three densities in Table II are recovered. At $n_{\text{grid}}=128$, the same splat produces a hollow shell instead: verified directly in job 833349 (ford_sweep_driver.py --config v3, run July 15, 2026; documented in docs/track1_v3_



Analytical model, not simulation output. $C_D=1.38$ (Smith, Modra & Felder 2019, 1:18 Yaris flume); $\mu=0.30$ (AR&R adopted, conservative); $m=1100$ kg; prism $4.3 \times 1.7 \times 1.47$ m over 0.75 of footprint.

Fig. 4. Analytical force-balance calculation, not simulation output. Left: lateral drag force versus flow velocity at four depths, against the maximum static friction limit (dashed). Right: the resulting critical velocity as a function of depth, with the band spanning $\mu = 0.30$ to 0.78 . Coefficients are pinned to primary sources: $C_d = 1.38$ [4], $\mu = 0.30$ [3], and the 0.78 upper endpoint of the shaded band [4], mass 1100 kg. The legend reports N directly; it is 0.00 kN at 0.30, 0.45, and 0.60 m. Those zeros are a real buoyancy result, not missing values: at 0.30 m the buoyant force reaches 16.135 kN against a weight of 10.791 kN, following from the effective plan area of 5.4825 m^2 , so N is zero and the friction limit μN vanishes with it. The right panel corroborates this independently, with critical velocity falling to zero at 0.201 m. That flotation depth follows from the model’s one remaining assumption, a solid prism over 0.75 of the bounding-box footprint. The hull encloses 3.5427 m^3 against a 10.7457 m^3 nominal prism ($4.30 \times 1.70 \times 1.47$ m), a fill factor of 0.33; against the mesh’s own 11.3533 m^3 extent it is 0.312. Either way the prism overstates displacement and floats the car at about 0.20 m, shallow against the 0.34 to 0.67 m reported for real cars and AR&R’s 0.30, 0.40, and 0.50 m still-water limits [3]. That bias runs toward predicting loss of traction too early, and is stated here rather than absorbed into the curves.

sweep_invalid_hollow_vehicle.md), particle count scaled $4.31 \times$ between the two resolutions rather than the $8 \times$ a solid body requires, and 0 of 60 test runs at $n_{\text{grid}}=128$ passed a density plausibility check (all three classes’ apparent density roughly doubled, $1.84\text{--}1.87 \times$, a resolution artifact, not a physical effect). Water resolution and vehicle solidity are governed by different length scales for this asset, raising grid resolution helps the former and silently breaks the latter. **Table II’s densities hold only at $n_{\text{grid}}=64$** and must not be read across to any higher-resolution run without re-solidifying the vehicle by a decoupled method first. This also settles, rather than merely defers, the question of testing 0.05 m depth: at $n_{\text{grid}}=64$ the material points sit about 0.074 m apart, which is wider than the 0.05 m being requested, so it seeds zero water particle layers for the pickup class, and at $n_{\text{grid}}=128$ the vehicle itself is invalid, so 0.05 m is not reachable with this asset at any grid setting tested.

The candidate fix this diagnosis points to, moving to a true watertight mesh rather than re-tuning the solidification pitch, is what the project adopted for vehicle representation tied to AR&R’s own classes. A watertight 2010 Toyota Yaris hull (yaris_coarse_v11_watertight.ply) replaces the single rescaled silhouette with one real geometry. It comes from the crash-validated finite-element model that the Center

TABLE II
VEHICLE CLASS DENSITY AT $n_{\text{GRID}}=64$ (ONLY VALID GRID SETTING)

Class	Density (kg/m ³)	In heuristic band?
Sedan	293.55	Yes
Pickup	242.12	Yes
SUV	308.13	No (2.7% over, left as-is)

Verified against
docs/track1_v3_sweep_invalid_hollow_vehicle.md. Values
are not meaningful at $n_{\text{grid}}=128$ (see body text).

for Collision Safety and Analysis publishes for NHTSA [19]. We use 1100kg as the baseline mass. CCSA reports two different weights for this model: the real car they measured weighed 1078 kg, and their finite-element version of that car weighs 1101 kg. We simulate the mesh and not the physical car, so 1101 kg is the figure that applies, and we round it to 1100 kg. That also matches the 1100 kg small-car test specification the model was built to meet. Vehicle class is then a mass override on that same geometry rather than three separate meshes (renders/yaris_render_s1/gates_both_scenarios.py): small_passenger at 1100 kg, large_passenger at 1609 kg, and large_4wd at 2337 kg. AR&R does not publish a single mass for each class. It publishes a kerb-weight bound: under 1250 kg for small passenger, over 1250 kg for large passenger, and over 2000 kg for large 4WD [3]. Each of our three masses falls inside the right bound, but they are representative values we chose, not numbers AR&R states. This resolves the class-mapping question in Fig. 3: AR&R’s classes do not map onto the sedan/SUV/pickup NHTSA/SAE classes used for the diagnostic asset in Table II at all; they are a separate three-mass study on one watertight hull, kept deliberately distinct from the silhouette-based diagnostic above. This mass-override asset and method, not the silhouette splat, is what any AR&R-class-linked simulation result in this paper is run on. The Yaris is also the small-passenger vehicle used in the full-scale traction testing of Smith, Modra and Felder [4] and in the scale-model work of Azhar, Pauwels and Bui [8], so the simulated geometry corresponds directly to the vehicle behind the empirical stability data this work compares against.

IV. RESULTS

A. Scene Reconstruction Status

Structure-from-motion reconstruction (COLMAP) is complete for one real scene (“Drain A”), confirmed via a populated sparse point cloud and camera pose set on TACC Lonestar6 scratch storage. A Gaussian splat was then trained from that reconstruction on Lonestar6 for 30,000 iterations, reaching a PSNR of 22.7 dB (SSIM 0.82, LPIPS 0.31) on held-out views with 1,147,694 Gaussians. It is not yet simulation-ready: monocular structure-from-motion recovers a scene only up to an unknown scale, no reference length in this scene has been measured to fix it, and the PhysGaussian bridge that would convert the splat into MPM particles is not built. A second

scene (“Drain B”) has not been captured or reconstructed. No result reported in this paper depends on either scene; the reconstruction stage is reported here as pipeline status, and every quantitative result below comes from synthetic geometry or the watertight hull.

B. Synthetic-Geometry Pilot Study

An early parametric comparison of L1 against a Genesis SPH solver (data/l2_results_from_wandb.csv) covers 9 unique (depth, velocity) conditions. L1 and L2 agreed at 5 of 9 (55.6%). The 4 divergence points, where L1 returned FORD but L2 returned NO-FORD, were (0.30 m, 1.5 m/s), (0.15 m, 1.5 m/s), (0.30 m, 1.0 m/s), and (0.30 m, 2.0 m/s). These figures supersede an earlier draft of this subsection, which reported 23 pairs, 16 divergence points, and 30.4% agreement. Those numbers do not match this dataset and were never traced to a source file, so we treat them as unverified and report only what is read directly from the CSV. A separate file, data/phase_space_results.csv, does contain 23 unique (depth, velocity) pairs, matching the older pair count, but it carries a single verdict column with no corresponding L1 value, so no agreement rate can be computed from it; 15 of its 31 rows additionally share a condition with another row under a different verdict, with three separate (0.30 m, 1.5 m/s) runs returning FORD, NO-FORD, and NO-FORD at three different displacements, so it is not cleanly enough run-labelled to use as it stands. Reconciling that file against an explicit L1 calculation remains outstanding.

A separate 4-point friction sweep at the (0.30 m, 1.5 m/s) divergence point (data/mu_sweep_results.csv) shows lateral displacement of 0.399, 0.3957, and 0.3953 m at coupling friction $\mu = 0.3, 0.5, 0.7$ respectively, essentially invariant across that range. At $\mu = 0.0$, displacement drops to 0.3283 m, a real difference, not noise. The invariance therefore holds across the $\mu = 0.3$ to 0.7 range and not from zero, a distinction an earlier draft of this work elided by describing the sweep as friction-invariant across $\mu = 0.0$ to 0.7. The qualitative point is unaffected: a scalar hazard measure cannot represent this directional force. The sweep is nonetheless four points at a single condition, and we do not generalize beyond it.

Fig. 5 plots that verified 9-point dataset directly. It is deliberately not extrapolated beyond those nine conditions, and the divergence structure it shows should be read as characterizing this pilot dataset rather than the full operating envelope.

C. Real-Simulation Sweep

These 17 runs were executed with the Warp-based MPM solver warpppm from kks32/mpm-engine (MIT), not with Genesis; Genesis is the engine the Fig. 1 pipeline is designed around and the solver behind the 9-condition pilot above, so the two should not be read as the same. The coupled sweep supporting AR&R-class-linked results is the gated inventory in data/all_runs_inventory.csv: 17 runs on the watertight Yaris hull (Section III-C), not track1_sweep_v2, which is superseded and excluded: that

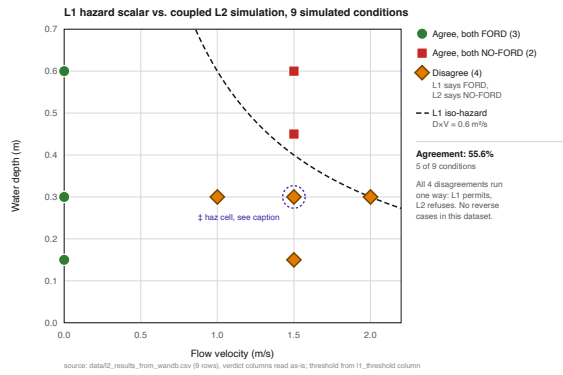


Fig. 5. L1 hazard scalar versus coupled L2 simulation, 9 simulated conditions from `data/l2_results_from_wandb.csv`. Agreement holds at 5 of 9 (55.6%): 3 conditions where both agree FORD, 2 where both agree NO-FORD. All 4 disagreements run one way, L1 permits and L2 refuses, with no reverse cases in this dataset. One flagged cell (marked †) has an `l1_haz_score` that disagrees with its own $D \times V$ product; the figure plots the consistent $D \times V$ value throughout and the discrepancy is noted rather than silently resolved.

earlier 36-run sweep used a rescaled box proxy of 1390 kg enclosing 4.7352 m³, against the real hull’s 3.5427 m³. Dividing the three masses by that hull volume gives 310, 454, and 660 kg/m³. All three sit below water at 1000 kg/m³, so all three configurations float once fully submerged, which is what a real car does. It comprises a 3×3 mass-by-resolution block (9 runs at `n_grid` 48/64/96, masses 1100/1609/2337 kg, one inside each of AR&R’s three kerb-weight bounds as described in Section III-C, which is a bound on kerb weight only and not class membership) fixed at depth 0.30 m and velocity 1.5 m/s, plus a three-point depth sweep and a five-point velocity sweep at `n_grid`=64. These three masses are applied as overrides to a single hull, so the block is a controlled mass-sensitivity study on one geometry rather than a comparison of three vehicles. The override changes mass but cannot change the hull. AR&R sorts vehicles on three things at once: length, kerb weight, and ground clearance [3]. Large passenger requires a length above 4.3 m and large 4WD above 4.5 m, whereas the Yaris hull measures 4.2826 m. Ground clearance was measured from the mesh at 0.1737 m, taken as the minimum z of a central longitudinal strip that excludes the wheels; the underbody has real structure rather than a flat reconstruction cap, with only 4.3% of strip vertices within 20 mm of that minimum. The unscaled hull therefore fails the length criterion for both upper classes at every mass tested, and its measured 0.1737 m clearance also exceeds the 0.12 m ceiling that small passenger requires. No AR&R class is satisfied on all three axes: small passenger passes length and kerb weight but fails clearance, large passenger passes kerb weight and clearance but fails length, and large 4WD passes kerb weight alone. The class labels carried in `data/all_runs_inventory.csv` are therefore assigned on kerb weight alone and are a labelled, reversible assumption rather than a demonstrated class membership. This matters for the verdict and not only for the label: for `g64_m1100` the realized depth of 0.2944 m at

1.5 m/s gives $D \times V = 0.4416 \text{ m}^2/\text{s}$, which exceeds the small-passenger limit of 0.30 m²/s and falls below the large-passenger limit of 0.45 m²/s, so the L1 verdict for the canonical run flips from NO-FORD to FORD on the class label alone, by a margin of 0.0084 m²/s. The floor-friction parameter passed to the solver’s `add_plane` call is held uniform at 0.55 across all 17 runs. That number is not arbitrary. Azhar et al. measured 0.55 as the wet friction of the rubber mat they used to stand in for a road, and checked it against a published range of 0.50 to 0.70 for tyres on wet asphalt [8]. Inside the solver, it still acts as a contact coefficient on a plane rather than as a tire model, so it should not be read as a measured tire-road value. It also does not match the assumption L1 was built on. AR&R’s thresholds were derived assuming a friction coefficient of 0.3, which Shand et al. call conservative but say the available data was not good enough to refine [3]. Our two levels are therefore not running at the same friction. The direction of that mismatch has been tested in part. The coupling-friction sweep in Section IV does not address it, because it varied the friction between vehicle and water, not between vehicle and ground. A separate, non-canonical study on the same `warpmpm` solver (a later driver script) varied the floor friction directly at a nominal depth of 0.30 m and velocity 1.5 m/s, and peak drift fell as floor friction rose in every case. At a grid spacing of 0.085 m (seven water layers), the 1100 kg Yaris hull drifted 0.96 m at 0.30 (one run) and 0.565 m at 0.55 (three runs), and a 2270 kg pickup hull drifted 0.163 m at 0.30 (one run), more than three times the 0.05 m detector, but 0.047 m at 0.55 (two runs from one initial condition), just under it. At a 0.136 m spacing (four water layers, as in the `n_grid`=64 runs at 0.30 m here, whose spacing is 0.147 m), the pickup drifted 0.206 m at 0.30 (one run) and still exceeded the detector at 0.55, at 0.087 to 0.092 m in all 17 repeat runs of that case. Higher floor friction therefore moves L2 toward FORD, so the mismatch works against, not toward, L2 refusing where L1 permits; whether it changes a given verdict depends on resolution. The finer-grid runs cited here all exceed the 10% particle-passthrough gate (10.2 to 11.2%), while every coarser run passes it, and the heavy vehicle has a different geometry, so these results establish the direction of the friction effect, not its size in this sweep. Rerunning the canonical sweep at 0.30 remains the direct test. Of the 17 canonical runs, 7 exceed a 10% particle-passthrough gate, reaching 15.9% at 3.0 m/s; these are flagged, not excluded. Water-layer depth resolution also varies across the block, from three layers at the coarsest grid to six at the finest, so the coarse-grid rows are the least trustworthy independent of their passthrough figure. We do not report an L1-versus-L2 agreement rate for this sweep. Doing so would require committing to the L2 drift threshold as a decision boundary, and Table I defines that threshold as a solver-internal numerical detector rather than an empirical criterion; until it is placed on an independent empirical footing, we report displacement magnitudes only. That fallback carries a caveat we state rather than absorb, because displacement magnitude is the least-converged quantity in the block. At 1100 kg it moves +87.8% from `n_grid`=48 to 64 and then −59.2% from 64

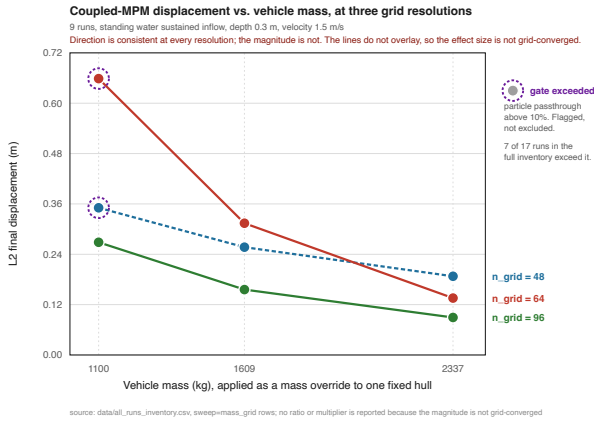


Fig. 6. The direct visual for the 3×3 mass-by-resolution block described in Section IV-C: coupled-MPM final displacement versus vehicle mass at three grid resolutions, from the nine `sweep=mass_grid` rows of `data/all_runs_inventory.csv`. One hull, three AR&R mass overrides, fixed depth 0.30 m and velocity 1.5 m/s. Displacement decreases with mass at every resolution, but the curves do not collapse onto one another and the 48- and 64-cell curves cross, so the effect size is not grid-converged; no ratio or multiplier is reported for this reason. Ringed markers exceed the 10% particle-passthrough limit and are flagged rather than excluded; 7 of the 17 runs in the full inventory exceed it.

to 96; at 1609 kg, +22.3% then -50.3% ; at 2337 kg it instead falls monotonically, -27.7% then -34.0% , so the sign of the resolution effect is not consistent across the three masses. Grid refinement is not expected to converge an instantaneous quantity, so this is not by itself evidence of a solver defect, but no convergence claim can rest on these magnitudes. The value is also route-dependent: for `g64_m1100` it is 0.658537 m read from the run summary against 0.637019 m recomputed from the stored particle rollout, a gap of 0.0215 m that is 3.3% of the first figure and 3.4% of the second. We therefore report these magnitudes descriptively and draw no ordering from them. The drift verdict itself is stable across the block: all nine runs cross the 0.05 m detector, and `g64_m1100` crosses it at frame 3 of 90. That stability is shown for these three grids only and should not be read as grid convergence of the verdict.

Fig. 6 shows the mass-by-resolution block. Displacement falls monotonically as mass rises at every one of the three grid resolutions tested, so the *direction* of the mass effect is consistent. The *magnitude* is not: the three curves do not overlay, and the 48- and 64-cell curves cross between 1609 and 2337 kg. The lightest-to-heaviest displacement ratio is non-monotonic in resolution across the three grids, so no single multiplier characterizes the mass effect and none is quoted here. Any mass-sensitivity claim from this work is therefore directional only, and treating it as a calibrated effect size would be unsupported.

Fig. 7 shows a single frame from the 1100 kg, `n_grid=64` member of that block.

V. CONCLUSIONS

The synthetic-geometry pilot demonstrates, as a proof of concept, that a scalar depth-velocity hazard measure can

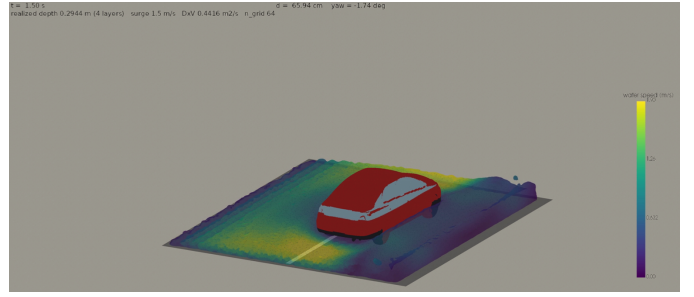


Fig. 7. One frame (45 of 90, $t = 1.50$ s) from run `g64_m1100` of the same `data/all_runs_inventory.csv` inventory: the watertight Yaris hull carried at 1100 kg (AR&R small passenger), `n_grid=64`, requested depth 0.30 m (realized 0.2944 m, four water layers), and inflow velocity 1.5 m/s, giving $D \times V = 0.4416 \text{ m}^2/\text{s}$. This is coupled-MPM solver output, not a Gaussian-splat reconstruction: the vehicle is the crash-validated finite-element hull carried as a rigid body, and the water surface is an isosurface reconstructed from the run's own MPM particles, coloured by water speed. The vehicle is drawn as the full watertight mesh obtained via the rigid-body transform, while the physics runs on the 8,905 material points that fill that hull. It is included to make the coupled simulation legible as a physical scene, not as a validated result. This run is one of the 7 of 17 that exceed the 10% particle-passthrough gate, at 10.7%, and the frame is mid-run, so the displacement printed in the frame is not the run's final reported value.

disagree substantially with a coupled physical simulation, and that the disagreement is not closed by coupling friction: at the one divergence point where friction was swept, no value in the tested range changes the verdict. Whether the pilot's disagreement is structural rather than a calibration artifact remains open, because the pilot ran on Genesis SPH rather than the `warpmpm` solver of the coupled runs; for the coupled solver, a non-canonical study in Section IV-C finds that higher floor friction reduces drift and so biases L2 toward FORD, with any effect on a given verdict depending on resolution. This supports treating abstraction selection as query-dependent rather than assuming any single fixed-fidelity model is correct everywhere in the operating envelope.

We scope that conclusion deliberately. It is a pilot-study result on synthetic geometry, not a claim about the real reconstructed scene: the Drain A splat is trained but not yet metrically scaled or coupled to the simulation, the real-scene sweep is not yet verified, and the vehicle representation carries the mesh and density limitations set out in Section III. This work establishes that the divergence exists in the tested regime, which motivates the orchestrator. Determining where the escalation boundary falls in a real scene requires a verified real-scene sweep, which remains future work.

VI. FUTURE WORK

We identified several concrete extensions, listed roughly in order of expected impact relative to effort.

- **Crash-validated vehicle geometry.** NHTSA/NCAC publishes free, government-funded, crash-validated finite-element vehicle models (e.g., a 2010 Toyota Yaris sedan [19] and a 2007 Chevrolet Silverado pickup) with real validated mass distribution, available in LS-DYNA keyword format and convertible to a simulation-ready collision mesh. We took the first step here. The Yaris hull in

Section III-C is one of these models, and every canonical AR&R-linked result in this paper runs on it. What remains is the second step: using a different published model for each class, so that the three classes differ in geometry as well as in mass.

- **Physical-viability audit dashboard.** A direct instantiation of PVWM’s “physically viable” criterion: automated checks for mass conservation, momentum conservation, MPM Jacobian/volume-ratio bounds, zero-penetration contact, and energy monotonicity, applied to every run rather than asserted informally.
- **Failure-mode classification with violation magnitude.** An explicit failure-mode hierarchy (stuck, slide, topple, float) reported with the magnitude by which a run crosses each threshold, rather than a binary verdict.
- **Graph Network Simulator (GNS) surrogate.** A lightweight GNS trained on the completed simulation sweep as a fast surrogate for untested (depth, velocity, vehicle-class) combinations, connecting the L0/L1/L2 ladder to a learned, still physically grounded fourth level.
- **A second independently-sourced real scene.** The Flooded Road Environments Dataset (FRED) captures on-road water hazards multi-modally, with camera, 64-beam LiDAR, and IMU/RTK-GNSS data from five locations recorded both during and after flooding, released with semantic labels in a KITTI-style format [20]. It is the natural second scene source for this pipeline: independent of our own capture, already labelled for water-hazard segmentation, and usable without a further field campaign.
- **DesignSafe-CI dataset publication.** Publish the validated real-scene dataset and simulation outputs with a DOI once the real-scene sweep is verified, rather than publishing the synthetic pilot data.

ACKNOWLEDGMENT

This work was supported by the National Science Foundation under NSF REU Site: Cyberinfrastructure Research for Societal Advancement (SCIPE-AI), Award #2447887, hosted at the Texas Advanced Computing Center (TACC), The University of Texas at Austin. The author thanks faculty mentor Dr. Krishna Kumar and research mentors Hassan Iqbal, Cheng-Hsi Hsiao, and Cristian Moran for their guidance throughout this project, and Rosalia Gomez for program coordination and support.

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